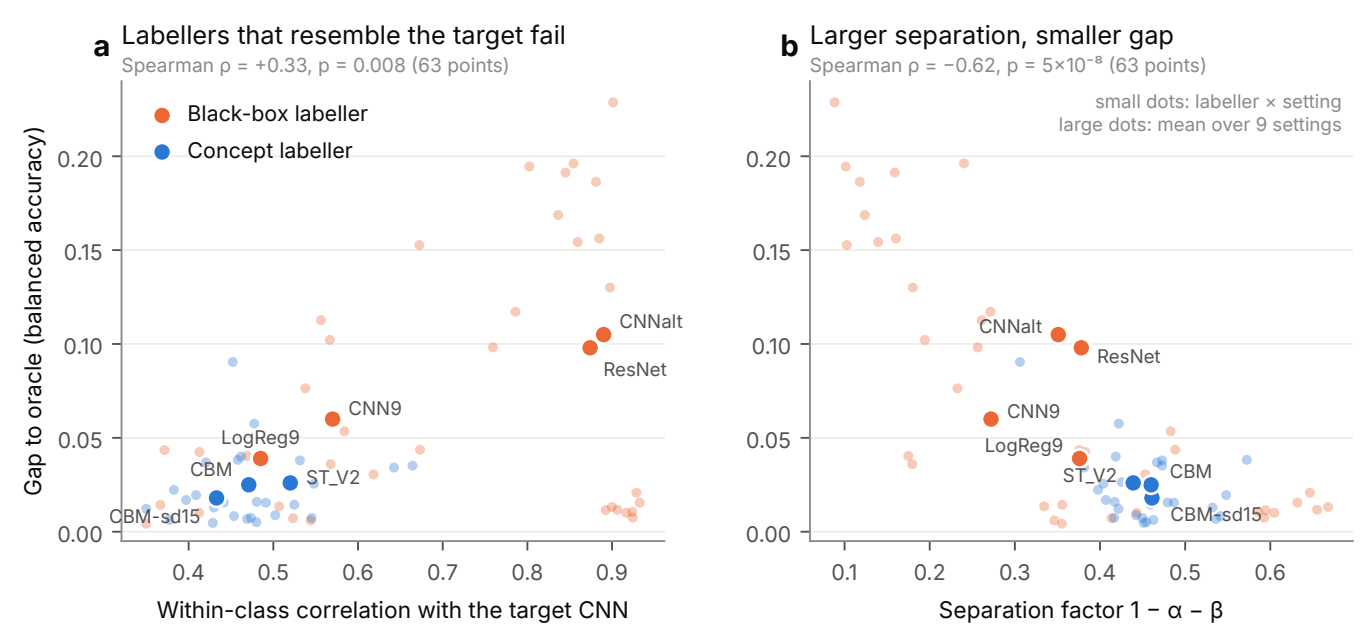
a
Labellers that resemble the target fail
Spearman ρ = +0.33, p = 0.008 (63 points)
Black-box labeller
Concept labeller
Gap to oracle (balanced accuracy)
0.00
0.05
0.10
0.15
0.20
0.4
0.5
0.6
0.7
0.8
0.9
Within-class correlation with the target CNN
CNNalt
ResNet
CNN9
LogReg9
CBM
ST_V2
CBM-sd15
b
Larger separation, smaller gap
Spearman ρ = −0.62, p = 5×10⁻⁸ (63 points)
small dots: labeller × setting
large dots: mean over 9 settings
0.00
0.05
0.10
0.15
0.20
0.1
0.2
0.3
0.4
0.5
0.6
Separation factor 1 − α − β
CNNalt
ResNet
CNN9
LogReg9
ST_V2
CBM
CBM-sd15